

Challenge: Identify Missing Data

Missing data is a common issue in real-world datasets, where some entries may be absent, incomplete, or recorded as "not available." Before you analyze or model your data, it is essential to identify where these missing values occur. Failing to address missing data can lead to inaccurate results, biased insights, or errors in downstream processing. Recognizing the presence and location of missing values is the first step in ensuring your data is clean and reliable for analysis.

```
import pandas as pd
import numpy as np

# Create a sample DataFrame with missing values
data = {
    "Name": ["Alice", "Bob", "Charlie", "David"],
    "Age": [25, np.nan, 30, 22],
    "City": ["New York", "Los Angeles", np.nan, "Chicago"],
    "Score": [85, 90, np.nan, 88]
}

df = pd.DataFrame(data)
print(df)
```